

11.1

Uvažujme reálnou matici A

**JURYS
INN**

$$A = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix}_{3 \times 4}$$

Jurys Inn Prague, Sokolovská 11, 186 00 Praha 8, Czech Republic
t +420 226 222 600 f +420 226 222 610 e info_prague@jurysinns.com
jurysinns.com

$$A^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix}_{4 \times 3}$$

Začneme tím, že spočítáme celý singulární rozklad matice $A^T = A^*$: $\text{rank} A = \text{rank} A^T = 3$

$$AA^T = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix}_{3 \times 4} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix}_{4 \times 3} = \begin{pmatrix} 7 & -2 & -2 \\ -2 & 7 & -2 \\ -2 & -2 & 7 \end{pmatrix}_{3 \times 3}$$

... vl.č. AA^T : 9, 9, 3

$$\text{vl.v. } AA^T: \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

... Sing. hod. $\sigma_1 \geq \sigma_2 \geq \sigma_3 > 0$ získáme jako $\sqrt{\text{vl.č. } AA^T}$
3 3 $\sqrt{3}$

[spočítáno pomocí W/A]

dle VĚTY o singulárním rozkladu: $A^T = UDV^*$

$$D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix}_{4 \times 3} = \text{diag}_{4 \times 3}(\sigma_1, \sigma_2, \sigma_3)$$

... $V = (\vec{v}_1 | \vec{v}_2 | \vec{v}_3)$ matice složená ze znormalizovaných
kolmých vl.v. AA^T

$$V = \begin{pmatrix} -1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 2/\sqrt{6} & 0 & 1/\sqrt{3} \end{pmatrix}_{3 \times 3}$$

pro AA^T , $\Pi_9 = \text{Span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\} \ni \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} \perp \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$$U = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -2/\sqrt{6} & 0 & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 0 & 1/\sqrt{3} \end{pmatrix}_{4 \times 4}$$

$$\Rightarrow \vec{v}_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}, \vec{v}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \vec{v}_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

... $U = (\vec{u}_1 | \vec{u}_2 | \vec{u}_3 | \vec{u}_4)$ $\leadsto \vec{u}_1, \vec{u}_2, \vec{u}_3$ spočítáme podle vzorce $\vec{u}_i = \frac{1}{\sigma_i} A^T \vec{v}_i$
 $\leadsto \vec{u}_4$... doplníme na ON bázi $\mathbb{R}^4$

$$\vec{u}_1 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix} \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} = \frac{1}{3\sqrt{6}} \begin{pmatrix} 0 \\ -6 \\ 3 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 0 \\ -2 \\ 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -2 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 0 & -2 & 0 & 2 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\vec{u}_2 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{3\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ -2 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ -1 \\ 3 \end{pmatrix}$$

$$\dots \vec{u}_4 = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\vec{u}_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix} \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$\Rightarrow$ další strana

Singulární rozklad A^T :

$$A^T = UDV^* = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{2}{\sqrt{6}} & 0 & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{2} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{1}{2} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix}$$

4×4 4×3 3×3

Singulární rozklad A :

$$(A^T)^T = A = VD^*U^* = \begin{pmatrix} -\frac{1}{\sqrt{6}} & -\frac{1}{2} & \frac{1}{3} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

3×3 3×4 4×4

Singulární rozklad $A^T A$:

$$A^T A = UDV^*VD^*U^* = UD^2U^* = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{2}{\sqrt{6}} & 0 & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{2} & 0 & \frac{1}{3} \\ \frac{1}{\sqrt{6}} & \frac{1}{2} & 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

4×4 4×4 4×4

Singulární rozklad AA^T :

$$AA^T = VD^*U^*UDV^* = VD^2V^* = \begin{pmatrix} -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

3×3 3×3 3×3

komplexní sdružení + transponování $*$ nad $\mathbb{R}$ odpovídá transponování T

Všetiváme toho, že $A = UDV^*$ je sing. rozkl. $\Leftrightarrow A^* = VD^*U^*$ je sing. rozkl.